Kathmandu University

Department of Electrical and Electronics Engineering

Master of Engineering in Communication Engineering Program

ECPG 535 Satellite Communications and Broadcasting

Test-I

$\frac{E_b}{N_0}$ and Modulation schemes

1. BPSK

Coherent detection of received Binary Phase Shift Keying (BPSK) can be written as,

$$\boldsymbol{y} = \boldsymbol{x} + \boldsymbol{n}$$

Where, amplitude $x \in \{-A, A\}$ and gaussian noise $n \sim N(0, \sigma^2) = N(0, \frac{N_0}{2})$ for a signal. In BPSK, minimum distance $d_{min} = 2A$ obtained from a constellation diagram. In Fig 1. probability of error in an overlapped region can be reduced into Q-function. For minimum error—with an optimum threshold zero, probability of error in detection when *-A* was sent is same as that of error when A was transmitted. Using Bayes theorem, for equiprobable probabilities of each symbol *-A* and A with same conditional probabilities, total error P_{BER} is equal to individual conditional probabilities. This can be expressed mathematically as,

$$P_{error|-A} = P_{error|A} = P_{BER} = \int_0^\infty \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(m+A)^2}{2\sigma^2}} d\,m$$

$$P_{BER} = \frac{1}{\sqrt{2\pi}} \int_{\frac{A}{\sigma}}^\infty e^{-\frac{k^2}{2}} d\,k$$

while Q-function is,

$$Q(n) = \frac{1}{\sqrt{2\pi}} \int_n^\infty e^{-\frac{y^2}{2}} d\,y$$

SNR per bit is an average energy required to represent a bit irrespective of modulation schemes written as $\frac{E_b}{N_0}$. For BPSK,

$$\frac{A^2}{\sigma^2} = \frac{2E_b}{N_0}$$

Therefore, $P_{BER} = Q\left(\sqrt{\frac{2E_b}{N_0}}\right)$

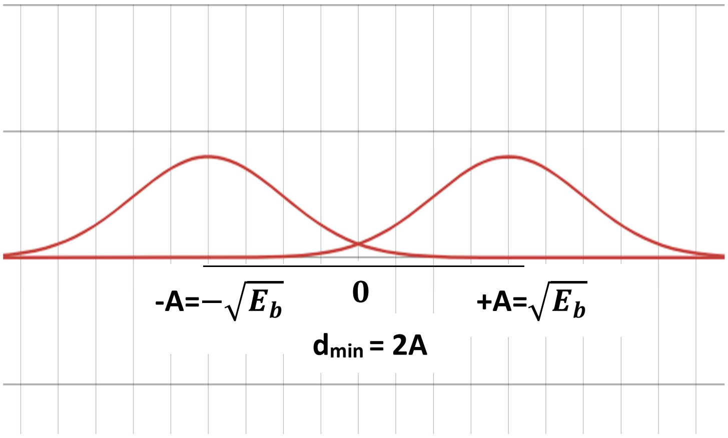

Figure 1 PDF of a signal in a coherent detection of BPSK modulated signal.

2. Modulated signals M-PSK, M-QAM

Using a constellation diagram approach, QPSK modulation is similar to BPSK modulation along in-phase and quadrature-phase component of the signal. So, bit error rate (BER) equals to that of BPSK.

For others signal, BER are equivalent to the values shown by the expression in table 1[1], [2].

Table 1 BER expression of different modulation schemes

Modulation	BER
BPSK	$Q\left(\sqrt{2\frac{E_b}{N_0}}\right)$
QPSK	$Q\left(\sqrt{2\frac{E_b}{N_0}}\right)$
M-PSK	$\frac{2}{\log_2 M} Q\left(\sqrt{2\frac{E_b}{N_0}\log_2 M}\sin\left(\frac{\pi}{M}\right)\right)$
Non-rectangular M-QAM	$\frac{4}{\log_2 M} Q\left(\sqrt{\frac{3\frac{E_b}{N_0}\log_2 M}{M-1}}\right)$

3. Plot

BER of different modulation scheme using approach discussed above is plotted in python code. Code is available in (github.com).

4. Observation and analysis

In coherent detection of BPSK and QPSK, signals affected by AGWN signal can be recovered with BER less than 10^{-6} if SNR per bit is greater or equal to 10.6 (~10.6 in theory). From graph, higher order M-array signals require larger SNR per bit due to nearer distance between symbols in a constellation diagram that are susceptible to interference by AGWN noise. Higher SNR in a link is obtained using amplification with the help of automatic gain control (AGC). This comes with the drawback of additional power consumption. A system designed for coherent detection of 32 M-array PSK requires $\frac{E_b}{N_0} \sim 21$. Same system can be designed using QPSK or BPSK with least BER and higher tolerable fade margin of 10.4 (~21-10.6) obtained from graph. In comparison with non-rectangular MQAM and MQPSK, QAM requires low SNR per bit due to different number of scaling amplitudes and phases as opposed to phase alone in MPSK.

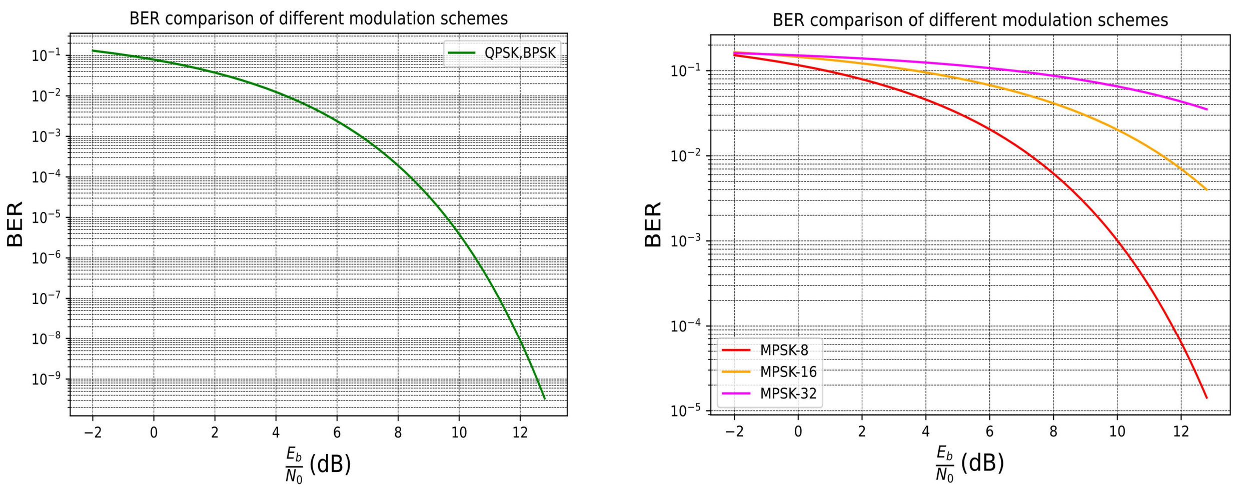

Figure 2 BER of BPSK, QPSK, and different levels of MPSK

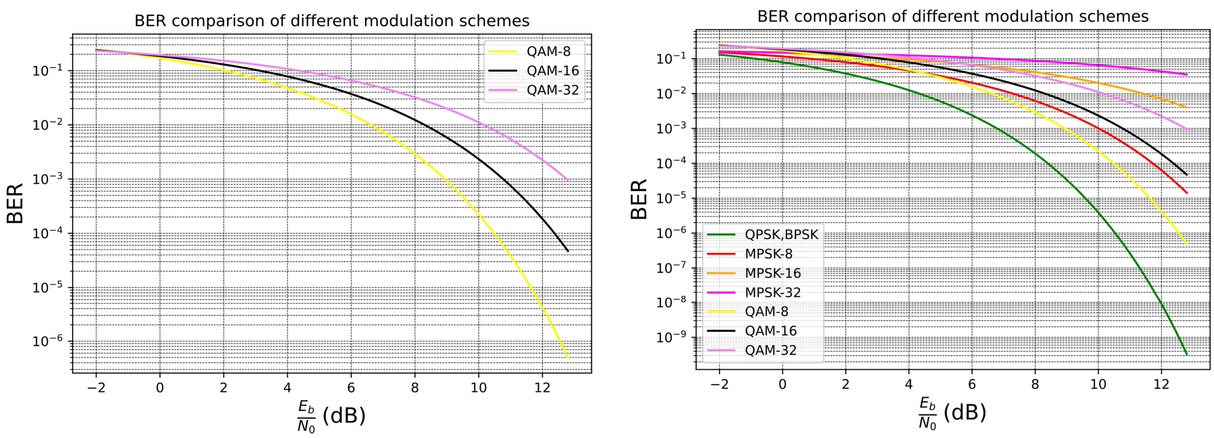

Figure 3 a. BER of different levels of non-rectangular QAM b. BER of different digital modulation schemes

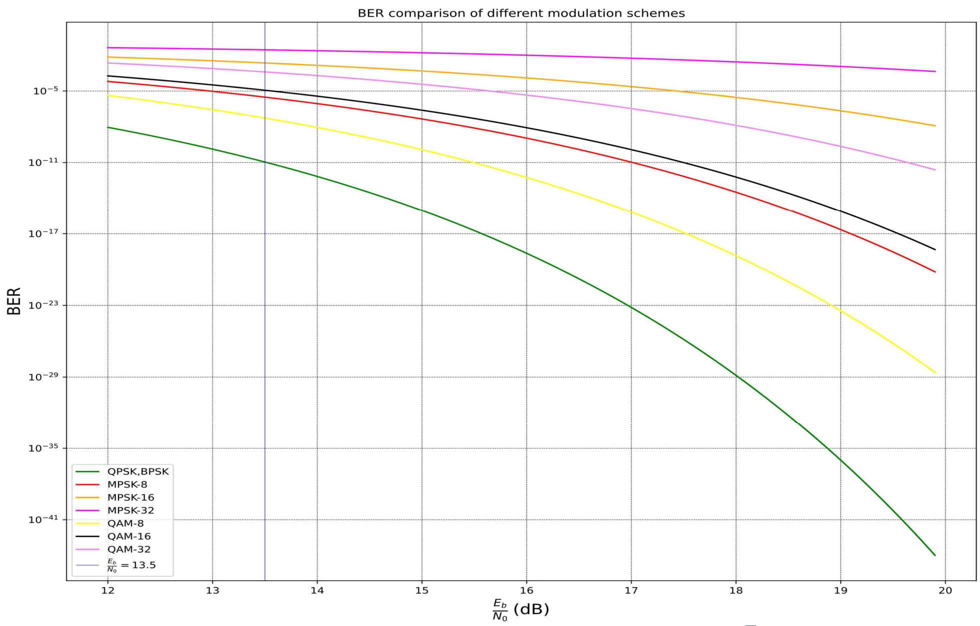

Figure 3 BER comparison for highest spectral efficiency when $\frac{E_b}{N_0}$ is 13.5

In coherent detection, SNR per bit depends on the minimum distance in a constellation diagram between adjacent symbol from a threshold and probability of error in symbol detection.

Digital modulation BPSK has symbol rate equals to bit rate R_b. The minimum bandwidth B_w to represent 1 bit is $\frac{1}{T}$ and number of bits per symbol is $\log_2(M = 1) = 1$, so, the spectral efficiency η is 1 bit/sec/Hz. QPSK has 2 bits per symbol and requires B_w equals to $\frac{1}{2T}$, therefore, spectral efficiency is 2 bit/sec/Hz. With $\frac{E_b}{N_0}$ obtained from graph, SNR can be calculated using Eq.6. Different modulation schemes SNR is calculated and tabulated in Table 2.

Table 2 Different modulation schemes SNR

Modulation	$\frac{E_b}{N_0}$ **(in dB)**	B_w	$log_2(M)$	η (Rate /Hz)	**SNR (in dB)**
BPSK	10.6	$\frac{1}{T}$	1	1	10.6
QPSK	10.6	$\frac{1}{2T}$	2	2	13.6
MPSK-8	14	$\frac{1}{3T}$	3	3	18.7
MPSK-16	17.4	$\frac{1}{4T}$	4	4	23.4
MPSK-32	21	$\frac{1}{5T}$	5	5	27.9
MQAM-8	12.8	$\frac{1}{3T}$	3	3	17.5
MQAM-16	13.6	$\frac{1}{4T}$	4	4	19.6
MQAM-32	15.6	$\frac{1}{5T}$	5	5	22.5

By the Shannon-Hartley theory, channel capacity C is the maximum information rate that can be transmitted over an AGWN channel is a function of bandwidth B and related to SNR as,

$$C = B\ log_2(1 + SNR) \qquad (7)$$

Eq. (7) can be reduced and expressed as,

$$\frac{C}{B} = log_2(1 + SNR) \qquad (8)$$

The ratio $\frac{C}{B}$ is called spectral efficiency η over an AGWN channel and its mathematical relationship with $\frac{E_b}{N_0}$ can be obtained solving Eq. 8 when C equals to R_b. This is written as,

$$\frac{E_b}{N_0} = \frac{2^{\frac{C}{B}} - 1}{\frac{C}{B}} \qquad (9)$$

Using Eq. (8) spectral efficiency for different modulation scheme is calculated in Table 3.

Table 3 Spectral efficiency of different modulation schemes in AGWN channel

Modulation scheme	$\frac{E_b}{N_0}$ (in dB)	Rate/Hz	Rate/Hz (in dB)	SNR (in dB)	SNR (in decimal)	Spectral efficiency
BPSK	10.6	1	0	10.6	11.4	3.6
QPSK	10.6	2	3.01	13.6	22.9	4.5
MPSK-8	14	3	4.7	18.7	75.3	6.2
MPSK-16	17.4	4	6.0	23.4	219.8	7.7
MPSK-32	21	5	6.9	27.9	629.4	9.3
MQAM-8	12.8	3	4.7	17.5	57.1	5.8
MQAM-16	13.6	4	6.0	19.6	91.6	6.53
MQAM-32	15.6	5	6.9	22.5	181.5	7.51

It can be inferred that among mentioned modulation schemes when SNR per bit is 21 in M-array PSK-32 signalling, spectral efficiency is maximum while BPSK has least spectral efficiency.

Conclusion

In this analysis, we find out that higher order modulation requires large SNR per bit to be unaffected by AGWN noises due to minimum distance between symbol in a constellation diagram. System designed for coherent detection of higher order signal has less fade margin compared to low order digital modulation. Despite this, higher order modulation exhibits better spectral efficiency, high data rate per bandwidth, which is important in bandwidth constraint mode.

Using graph in Fig. 3, for a link established with SNR per bit near 13.5, it is always preferable to use non MQAM-16 for higher spectral efficiency. If fade margin is to be

considered ~1.5 dB or 1.3 times the original SNR per bit, it is preferred to use MQAM-8.

Future work

Analysis doesn't include DVB-RCS, DVB-S2, DVB-T schemes which are essentially designed for satellite television broadcast. With reference to literatures, these has low SNR per bit and highest spectral efficiency near to attenable theoretical limit for AGWN channel, which will be discussed in future days if possible.

REFERENCES

[1] A. Goldsmith, *WIRELESS COMMUNICATIONS*. Cambridge University Press, 2005.

[2] V. Meghdadi, "BER calculation." Accessed: Aug. 28, 2024. [Online]. Available: https://www.unilim.fr/pages_perso/vahid/notes/ber_awgn.pdf